MESA SCHOOL OF BUSINESS

Data Foundations · Pre-Read

Session 9: Swiggy Instamart — Capstone

Big Mastery Question: Three tabs, no question. Find the decision, make it, defend it to a CXO.

Before you come to class

Read this once, slowly. There is NO new Sheets syntax this session — every formula you need, you already know from Sessions 1 through 8. What's new is combining them into the metrics a real founder's office tracks every week: cohort retention, funnel conversion, CAC, ROAS and LTV. This document explains what each metric MEANS and how it's built from formulas you already have. Every table below still shows real column letters and row numbers.

The company: Swiggy Instamart delivers groceries in minutes from local dark stores. For the capstone, you get THREE linked tables — orders, customers, and dark_store_master — and no guiding questions. Your job: find a real decision hiding in the data, build the case for it using the metrics in this document, and defend it in a 3-sentence memo, exactly the way you would in a founder's office.

0. The Datasets You'll Be Working With

Sheet 1 — "Orders" (9 columns). A 6-column, 7-row slice, spanning June and July:

	A	B	C	E	G	H
1	order_id	date	city	customer_id	order_value	discount
2	SW8001	2026-06-05	Bengaluru	C601	480	40
3	SW8002	2026-06-18	Bengaluru	C602	260	0
4	SW8007	2026-06-25	Delhi	C604	200	0
5	SW8003	2026-07-02	Bengaluru	C601	390	20
6	SW8004	2026-07-10	Mumbai	C603	610	50
7	SW8005	2026-07-15	Bengaluru	C602	150	0
8	SW8006	2026-07-22	Mumbai	C603	240	0

Column A = order_id, B = date, C = city, E = customer_id, G = order_value, H = discount. (D = dark_store_id, F = items, I = delivery_min, not shown.)

Sheet 2 — "Customers" (a separate tab, 7 columns). All 4 customers referenced above:

	A	B	C	E	G
1	customer_id	signup_date	city	total_spend	acquisition_channel
2	C601	2026-06-05	Bengaluru	870	Paid Social
3	C602	2026-06-18	Bengaluru	410	Organic

4	C603	2026-07-10	Mumbai	850	Referral
5	C604	2026-06-25	Delhi	200	Organic

On the Customers sheet: A = customer_id, B = signup_date, C = city, E = total_spend, G = acquisition_channel. (D = orders_count, F = last_order_date, not shown.) Notice C604 signed up in June and never ordered again after their first order.

1. The Cohort Retention Grid

Real world reference:

Imagine a gym tracking every batch of members who joined in the same month as one "class" — the "June class," the "July class" — and then checking, month after month, how many of each class are still showing up. That's a cohort: a group defined by WHEN they started, tracked over time.

In Swiggy Instamart's world:

Group customers by their SIGNUP month using EOMONTH (Session 6) — that's their cohort. Then, for each following month, use COUNTIFS (Session 3) to check whether each cohort placed at least one order in that month. The result is a grid that shows retention decaying (or holding up) over time — one of the most-watched charts in any startup.

The recipe (all formulas you already know):

```
Step 1 — Cohort Month:  =EOMONTH(signup_date, 0) Step 2 — Retained in a later month?:   =COUNTIFS(Orders!customer_id_range, customer_id, Orders!date_range, ">="&month_start,             Orders!date_range, "<="&month_end) > 0
```

See it in the table:

	A	B	C
1	**Cohort (signup month)**	**Cohort size**	**% retained in month 2**
2	June 2026	3	67%
3	July 2026	1	— (too early to tell)

June's cohort (C601, C602, C604): 2 of 3 ordered again in July (C601 and C602 did; C604 didn't) → 67% retained. This one grid answers "are we keeping customers?" far better than any single total ever could.

2. Funnel Conversion %

Real world reference:

Picture water poured into a funnel — a lot goes in the wide top, but only some of it makes it out the narrow bottom. Each narrowing loses a bit. Funnel conversion is simply: what PERCENTAGE made it from one stage to the next?

In Swiggy Instamart's world:

Every customer passes through stages: Signed Up → Placed 1st Order → Placed 2nd Order (became a repeat customer). COUNTA and COUNTIFS (both from Session 3) count how many people are at each stage — and dividing one stage's count by the previous stage's count gives you the conversion rate between them.

The recipe:

```
Conversion % (Stage A → Stage B) = COUNT at Stage B / COUNT at Stage A
```

See it in the table:

	A	B	C
1	**Funnel Stage**	**Customers**	**Conversion from prior stage**
2	Signed Up	4	—
3	Placed 1st Order	4	100%
4	Placed 2nd Order (repeat)	2	50%

All 4 signups placed a first order — a strong top of funnel. But only 2 of those 4 came back for a second order — the 50% drop is exactly where a founder would want to dig in.

3. CAC — Customer Acquisition Cost

Real world reference:

If you spend ₹500 on fishing bait and catch 5 fish, each fish effectively "cost" you ₹100 to catch. CAC is that exact idea applied to customers instead of fish — how much did it cost, on average, to win ONE new customer?

In Swiggy Instamart's world:

The acquisition_channel column tells you WHICH channel brought in each customer — but the spend itself usually lives in a separate marketing ledger, not in this table. Combine the two: count new customers per channel with COUNTIF (Session 3), then divide that channel's total spend (given by the marketing team) by that count.

The recipe:

```
CAC = Total spend on a channel / New customers acquired via that channel
```

See it in the table:

	A	B	C	D
1	**Channel**	**New customers (COUNTIF)**	**Spend (from marketing)**	**CAC**
2	Paid Social	1	₹4,000	=4000/1 → ₹4,000
3	Organic	2	₹0	=0/2 → ₹0

Organic customers effectively cost nothing to acquire directly — but that doesn't mean they're free overall (brand, content and SEO work still cost something upstream).

4. ROAS — Return on Ad Spend

Real world reference:

If you put ₹1 into a vending machine and get a ₹3 snack out, that's a great trade. ROAS asks the same question of marketing spend: for every rupee spent, how many rupees came back in revenue?

In Swiggy Instamart's world:

Take the revenue generated (SUMIF on total_spend, grouped by channel — Session 3) and divide it by that channel's ad spend. A ROAS above 1 means the channel is paying for itself already; below 1 means it hasn't yet —

which is completely normal early on, since a customer's value builds up over many future orders, not just their first.

The recipe:

```
ROAS = Revenue generated by a channel / Spend on that channel
```

See it in the table:

	A	B	C	D
1	Channel	Revenue so far (SUMIF)	Spend	ROAS
2	Paid Social	₹870	₹4,000	=870/4000 → 0.22

A ROAS of 0.22 looks alarming in isolation — but C601 is only 1 month into their relationship with Swiggy Instamart. This is exactly why LTV, not day-one revenue, is the real test of whether a channel works.

5. LTV — Customer Lifetime Value

Real world reference:

If a friend asks to borrow ₹100 once, that's a small ask. If they're going to be your roommate for 3 years and split every bill, their true "value" to your finances is everything across that whole relationship — not just today's ₹100. LTV is that same idea applied to a customer: their value across the ENTIRE relationship, not just one order.

In Swiggy Instamart's world:

The simplest LTV proxy you already have is total_spend — the running total of everything a customer has spent so far. It's not their FINAL lifetime value (they might order many more times), but it's a live, honest running estimate you can track and watch grow.

The recipe:

```
LTV (running proxy) = total_spend to date (a more advanced version: AOV × expected number of future orders)
```

See it in the table:

	A	G	E
1	customer_id	acquisition_channel	total_spend (LTV so far)
2	C601	Paid Social	870
3	C602	Organic	410

6. LTV:CAC Ratio

Real world reference:

Back to the fishing example: if bait costs ₹100 and the fish you catch is worth ₹300, that's a great trade — a 3:1 ratio. If the fish is only worth ₹50, you're losing money on every catch. LTV:CAC is exactly that comparison, applied to customers: is what they're worth bigger than what it cost to get them?

In Swiggy Instamart's world:

This single ratio is often the number that decides whether a channel gets MORE budget or gets cut entirely. A commonly cited healthy benchmark is 3:1 or higher — meaning a customer is worth at least 3 times what it cost to acquire them. Below 1:1 means the channel is currently losing money per customer.

The recipe:

```
LTV:CAC = LTV / CAC
```

See it in the table:

	A	B	C	D
1	**Channel**	**LTV (so far)**	**CAC**	**LTV:CAC**
2	Paid Social	₹870	₹4,000	=870/4000 → 0.22 (underwater, still early)

0.22 is far below the 3:1 benchmark — but remember, LTV here is only ONE MONTH of spend for a brand-new customer. The real question for your capstone: is this ratio climbing over time, or stuck? That trend is the decision hiding in the data.

Coming to class

Every formula in this document — EOMONTH, COUNTIFS, SUMIF, plain division — you already know. The capstone is entirely about combining them into metrics that matter, finding the one decision worth making, and defending it in three sentences a founder could act on immediately.